Case Study – VII-1

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Activity Link : https://asnm-iitkgp.vlabs.ac.in/List%20of%20experiments.html

Course Title & Course Code: Electric Circuit Analysis (3EP02)
Class: Second Year Electrical (Electronics & Power) Engineering
Semester: III
Name of the Course Teacher: Prof. V. S. Karale
Title of the Innovative Practice: Theorem-Based Concept Reinforcement through Virtual and Experimental Simulations

Objectives/Goals of the Practice

The key objective of this innovative practical teaching approach is to enhance conceptual understanding and analytical thinking in Electric Circuit Analysis through **virtual verification of network theorems** and **parameter evaluation of two-port networks** using simulation tools and hands-on circuit building.

Specific Objectives:

- To strengthen students' understanding of fundamental network theorems through practical verification.
- To develop the ability to correlate theoretical analysis with experimental results.
- To promote use of virtual lab environments (e.g., Multisim, PSPICE, or NI ELVIS) for simulation-based learning.
- To provide hands-on exposure to analyzing two-port networks and verifying parameter interrelations.

Use of Appropriate Methods

To achieve the learning objectives, the following methodology was adopted:

- **Simulation-Based Learning:**
  Each theorem was first introduced theoretically, followed by its simulation in software like **Multisim** or **PSPICE**. Students simulated the given circuit and verified the outcomes.
- **Stepwise Experimental Procedure:**
  Students followed structured steps to verify:
  1. Norton's Theorem
  2. Thevenin's Theorem
  3. Superposition Theorem
  4. Tellegen's Theorem
  5. Millman's Theorem
  6. Compensation Theorem

7. Determination and verification of parameters of a two-port network (Z, Y, h, T, T^{-1})

- **Comparative Analysis:**
  Simulation results were compared with theoretical calculations, helping students identify errors and refine their approach.
- **Interactive Feedback and Peer Discussion:**
  Students were encouraged to discuss their observations in small groups and submit comparative reports showing theoretical vs simulated values

Significance of Results

- **Conceptual Clarity:**
  The hands-on and visual approach improved comprehension of abstract theorems.
- **Analytical Skill Development:**
  Students learned how to simplify complex circuits and verify relationships between voltage and current using equivalent circuits.
- **Increased Accuracy:**
  Simulations reduced experimental errors and allowed precise measurements and parameter evaluation.
- **Inter-Parameter Understanding:**
  The study of two-port networks helped students understand how different parameter sets relate and convert into each other.

Effective Presentation

- Theoretical notes and step-by-step procedures were provided for each experiment.
- Video demonstrations were made available for students facing difficulties in simulation setup.
- Students submitted observation tables, screenshots of simulation results, and interpretation of errors.
- A comparative format of "Theory vs Simulation" was enforced in lab journals for critical thinking development.

Photo of the activity

Program Outcomes (POs) Mapped:

- **POs:** PO1, PO2, PO3, PO4, PO5, PO12
- **PSOs:** PSO1

Reflective Critique

Faculty noted that the **visual nature of circuit simulation** significantly improved the students' ability to grasp circuit behavior and the underlying logic of theorems.

Faculty Suggestions:

- Prof. G. N. Bonde emphasized the incorporation of **simulation-based assignments** to further develop independent learning.
- It was also recommended to include **hardware verification for selected theorems** in the upcoming semester for comparison with simulation.

Student Feedback:

- Students appreciated being able to **"see" current and voltage changes** during various stages of verification.
- Most students found Millman's and Compensation Theorems easier to understand after simulation.
- Students suggested integrating Arduino or basic sensors to see practical applications of these theorems.

Evidences of Success

- **90%+ submission rate** with correctly simulated results and aligned theoretical calculations.
- **Improved performance** in unit tests related to circuit theorems and network analysis.
- Students demonstrated an interest in more advanced circuit simulations and opted for minor projects involving network reduction and load matching.

Challenges Faced During Implementation

- Some students lacked prior exposure to simulation tools, requiring an initial orientation session.
- Limited system availability in labs necessitated batch-wise practice and teamwork.
- Ensuring every student had equal hands-on time during group simulations was challenging.
- A few theorems like Tellegen's required deeper mathematical explanation, taking extra teaching time.

Peer Review & Critique Form Link： https://forms.office.com/r/S77pz063xi